Advanced Computer Systems Engineering Laboratory (ENCS5150)

TODO 1 – Visual simulation of a Hash Table

Overview

Implement a **visual simulation of a Hash Table with Separate Chaining** with a fixed number of buckets and support for efficient insert, delete, and search/display operations, The app should visually show how keys are distributed across buckets in real time.

Core Features

- **INSERT(key, value):** Add a new key, value pair to the hash table.
- **DELETE(key):** Remove an existing key, value pair from the hash table.
- **SEARCH(key):** Find a key and display its value and bucket index.
- **Visual Table Display:** Real-time visualization of all buckets and their stored entries.
- **Fixed Table Size:** Maximum of 4 buckets.

Technical Implementation

- **MainActivity:** Visual hash table representation and operation controls.
- I**nsertActivity:** Interface for adding key–value pairs.
- **DeleteActivity:** Interface for deleting entries by key.
- **HashTable Class:** Fixed bucket count = 4.

Data Structures Required

- **Array / ArrayList of buckets :** fixed-size table 4 buckets.**.**
- **Linked List in each bucket:** for collision handling using chaining**.**
- **Hash Function:** maps key to bucket index.
- **Combination allows: average O(1) for insert, delete, search**

UI Requirements

MainActivity	InsertActivity	DeleteActivity
• Real-time visualization of hash table • 4 fixed buckets shown on screen • Each bucket displays its chain of key value pairs • Navigation buttons to INSERT, DELETE screens	• Form with key and value input fields • Submit button to insert entry then navigate to the main activity	• Input field for key • Delete button to remove entry • Status message for deleted / not found • Return button to main activity

Deadline: 30/3/2026 Midnight (Send it before 31/3/2026)

Example Scenario

1. Initial State

Bucket 0: empty
Bucket 1: empty
Bucket 2: empty
Bucket 3: empty

2. Insert (10, "A")

10 % 4 = 2
Bucket 2 → (10, A)

3. Insert (6, "B")

6 % 4 = 2
Bucket 2 → (10, A) → (6, B)

4. Insert (7, "C")

7 % 4 = 3
Bucket 3 → (7, C)

5. Delete 6

Bucket 2 → (10, A)

Final visual state

Deadline: 30/3/2026 Midnight (Send it before 31/3/2026)

Notes

- You can create any design as long as it is clear, easy to understand, and follows any mentioned specifics in the **"UI Requirements"** section above.
- The name of the application must be: ID_FirstName_LastName
- Use **Pixel 3a XL device with API Level 28 (Graphic = Software)**.
- The **minimum SDK** should be **API 28**.
- This TODO is **individual work**. Cheating or using LLM Models will result in **0 marks**.
- **No late submissions whatsoever**. If the deadline is about to arrive, submit your work as it is.

Marks Distribution (Total: 10 Marks)

Component	Description	Marks
Functionality	Implementation of all required features; buttons, game logic, navigation, and correct handling of user actions	4
Error-Handling & Stability	Proper handling of invalid inputs, prevention of crashes, smooth app behavior under different scenarios	2
Design & User Interface	Clear, visually appealing layout, adherence to color schemes, hints, and layout requirements	2
Usability & User Experience	Ease of use, intuitive navigation, responsiveness, and consistency between activities	1
Documentation & Naming Conventions	Correct app name format (`ID_FirstName_LastName`), proper project structure, meaningful variable names, comments …	1

Submission Requirements

1. **Project.zip file** (Size in KB)
    - From Android Studio: File → Export → Export to Zip File
2. **app-debug.apk file** (Size in MB)
    - From Android Studio: Build → Build Bundle(s) / APK(s) → Build APK(s)
    - You will find the APK file under: app\build\outputs\apk\debug\app-debug.apk

Send both the APK file and the ZIP file as a reply to my message.

Deadline

30/3/2026 Midnight (Send it before **31/3/2026**)